A2.3 Viruses (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **STATE** three features shared by all viruses. [A2.3.1] [3]

2 **NAME** the enzyme that HIV uses to make DNA from its RNA. [A2.3.2] [1]

3 **DEFINE** prophage. [A2.3.4] [1]

4 **STATE** two structural differences between a virus and a bacterium. [A2.3.1] [2]

5 **STATE** two shapes of viral capsid, with an example of each. [A2.3.2] [2]

Section B: Understanding

15 MARKS

6 **EXPLAIN** how an enveloped virus such as HIV gets its envelope. [A2.3.2] [2]

7 **OUTLINE** the stages of the lytic cycle of bacteriophage lambda. [A2.3.3] [4]

8 **EXPLAIN** why phage lambda cannot reproduce outside a bacterial cell. [A2.3.3] [2]

9 **COMPARE** the lytic and lysogenic cycles. [A2.3.4] [3]

10 **EXPLAIN** why viruses are thought to have arisen more than once. [A2.3.5] [2]

11 **EXPLAIN** why HIV infection is treated with a combination of drugs. [A2.3.6] [2]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [A2.3.3, A2.3.4]

A suspension of phage lambda was diluted in steps of ten. Then 0.1 cm^3 of each dilution was spread on a dish covered with a thin layer (lawn) of *E. coli*. After a day, clear spots called plaques were counted.

DILUTION	10^{-4}	10^{-5}	10^{-6}	10^{-7}
PLAQUES	too many to count	312	34	3

a EXPLAIN how a single phage produces one clear plaque. [1]

b SUGGEST why the 10^{-6} dish gives the most reliable estimate. [1]

c CALCULATE the number of phages per cm^3 in the undiluted suspension. [2]

d 0.1 cm^3 of undiluted suspension was added to a culture of 5.0×10^7 bacteria. The multiplicity of infection (MOI) is the number of phages per bacterium. **Calculate** the MOI. [2]

Phage lambda was then added to bacteria growing in a rich or a poor medium. The percentage of infected bacteria that became lysogens (carrying a prophage) was recorded: rich medium 8%, poor medium 46%.

e DESCRIBE the effect of the medium on lysogeny. [1]

f SUGGEST why the lysogenic cycle is an advantage to the phage when its hosts grow poorly. [2]

g The lysogens were exposed to UV light. PREDICT what would happen to the number of free phages, with a reason. [1]

Section D: Longer answer

6 MARKS

13 EXPLAIN why influenza viruses evolve rapidly, and the consequences for controlling flu. [A2.3.6] [6]